

A closed-form model of granularity, sequential clearing, and bid shape Reduced two-tranche schedules in DA and IDA under MTU60 vs MTU15

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Abstract

We extend the residual-monopolist linear-demand framework of the research proposal (Ito and Reguant 2016 reduced form) with a two-tranche reduced schedule in the spirit of Csereklyei and Khezzr (2024), under a uniform distribution for the within-hour fundamental. The model has closed-form solutions for the per-quarter Cournot clearing prices, the optimal two-tranche ladder under ex-ante demand uncertainty, and the storage arbitrage value. Three derived propositions: *(P1)* the cross-quarter price dispersion D_{price} widens under MTU15 with magnitude $\sigma_\mu/(2b^{DA})$; *(P2)* the optimal two-tranche ladder spreads with the half-range of the ex-ante demand distribution, $\Delta p^* = \Delta_\mu/(2b^{DA})$, which is hour-class specific and widens the per-curve σ_p critical-flat DiD by $(\Delta_C - \Delta_F)/(4b^{DA})$; *(P3)* pump-storage critical-hour arbitrage value scales linearly in Δ_μ/b^{DA} . The day-ahead cleared-volume scale-up under MTU15 is rationalised by importing the Csereklyei–Khezzr (2024) withholding mechanism without re-deriving it. The sequential DA-IDA structure is preserved throughout via the proposal’s arbitrage condition.

1 Setup

Time structure. A representative hour is divided into Q equal-length subperiods (*quarters*) indexed $q \in \{1, \dots, Q\}$. The reform changes Q : under MTU60 the relevant trading granularity is hourly ($Q = 1$), under MTU15 it is quarter-hourly ($Q = 4$). We label the three regimes *DA60/ID60*, *DA60/ID15* and *DA15/ID15*, following the proposal.

Residual demand. A residual monopolist faces a linear residual demand in each subperiod, on each side (day-ahead DA and intraday auction IDA):

$$q_q^{DA} = A_q - b^{DA} p_q^{DA}, \quad q_q^{ID} = b^{ID} (p_q^{DA} - p_q^{ID}), \quad b^{DA} > b^{ID} \geq 0. \quad (1)$$

The DA intercept is $A_q = \bar{A} + \mu_q$, with $\bar{A} > 0$ the constant hourly mean and μ_q a zero-mean within-hour shifter.

Distributional assumption. For closed-form tractability we take μ_q to be *uniformly distributed* on the symmetric support $[-\Delta_\mu, +\Delta_\mu]$ across quarters within an hour:

$$\mu_q \sim \text{Uniform}([-\Delta_\mu, +\Delta_\mu]), \quad \mathbb{E}[\mu_q] = 0, \quad \text{Var}(\mu_q) = \frac{\Delta_\mu^2}{3}. \quad (2)$$

We denote the resulting standard deviation by $\sigma_\mu \equiv \Delta_\mu/\sqrt{3}$. The half-range Δ_μ is hour-class specific:

$$\Delta_\mu = \begin{cases} \Delta_C & \text{if the hour is } \textit{critical} \text{ (morning ramp or evening peak),} \\ \Delta_F & \text{if the hour is } \textit{flat} \text{ (overnight),} \end{cases} \quad \Delta_C \gg \Delta_F.$$

Cost and reduced bid schedule. The firm has constant marginal cost c and capacity K . Rather than a full supply function, the firm posts a *two-tranche* ladder per delivery product (one ladder per period):

$$\text{Bid}_q = \{(p_q^L, k_q^L), (p_q^H, k_q^H)\}, \quad p_q^L \leq p_q^H, \quad k_q^L + k_q^H = K. \quad (3)$$

The empirical bid-shape moments are direct functions of (3):

$$\sigma_p = \sqrt{w^L w^H} (p_q^H - p_q^L), \quad N_{\text{eff}} = ((w^L)^2 + (w^H)^2)^{-1}, \quad w^L = k_q^L/K, \quad w^H = k_q^H/K. \quad (4)$$

Granularity matching. The number of distinct ladders the firm posts per hour is set by market design:

Regime	# DA ladders/hour	# IDA ladders/hour
DA60/ID60	1	1
DA60/ID15	1	4
DA15/ID15	4	4

2 Equilibrium prices and tranche structure (closed form)

2.1 Per-quarter Cournot price under MTU15

Under MTU15 the DA stage clears each quarter separately. The firm maximises $\pi_q^{DA} = (p_q^{DA} - c) q_q^{DA} = (p_q^{DA} - c)(A_q - b^{DA} p_q^{DA})$, with first-order condition $A_q - 2b^{DA} p_q^{DA} + b^{DA} c = 0$. This yields the per-quarter Cournot price:

$$p_q^{DA} = \frac{A_q + b^{DA} c}{2b^{DA}} = \frac{\bar{A} + b^{DA} c}{2b^{DA}} + \frac{\mu_q}{2b^{DA}} \quad (5)$$

with associated cleared quantity $q_q^{DA} = (A_q - b^{DA} c)/2$.

2.2 Hourly Cournot price under MTU60

Under MTU60 the DA stage clears once per hour at the hour-aggregate residual demand. The hour-aggregate intercept is $A_h = \frac{1}{Q} \sum_{q \in h} A_q = \bar{A} + \frac{1}{Q} \sum_q \mu_q$. Because the μ_q are mean-zero, in expectation $A_h = \bar{A}$; in any single hour the realised A_h is $\bar{A} + \tilde{\mu}$ with $\tilde{\mu} = \frac{1}{Q} \sum_q \mu_q$ of variance $\text{Var}(\tilde{\mu}) = \sigma_\mu^2/Q$. The hourly Cournot price is

$$p_h^{DA} = \frac{\bar{A} + b^{DA} c}{2b^{DA}} + \frac{\tilde{\mu}}{2b^{DA}}. \quad (6)$$

2.3 Closed-form cross-quarter price dispersion

Cross-quarter SD of the clearing price. Define

$$D_{\text{price}} \equiv \text{sd}_q(p_q^{DA}), \quad (7)$$

the standard deviation of the four quarter clearing prices within an hour.

Under *MTU60* all four quarters of an hour share the same hourly clearing price p_h^{DA} , so $D_{\text{price}}^{\text{MTU60}} = 0$ mechanically. Under *MTU15* we substitute (5) into (7):

$$D_{\text{price}}^{\text{MTU15}} = \text{sd}_q \left[\frac{\bar{A} + b^{DA} c}{2b^{DA}} + \frac{\mu_q}{2b^{DA}} \right] = \frac{\sigma_\mu}{2b^{DA}}. \quad (8)$$

Proposition 1 (Cross-quarter price dispersion widens under MTU15). *Under the uniform distribution (2),*

$$D_{price}^{MTU15} - D_{price}^{MTU60} = \frac{\sigma_\mu}{2b^{DA}} = \frac{\Delta_\mu}{2b^{DA}\sqrt{3}} > 0,$$

and the (critical – flat) differential of this object is strictly positive:

$$\left(\frac{\sigma_C}{2b^{DA}} \right) - \left(\frac{\sigma_F}{2b^{DA}} \right) = \frac{\sigma_C - \sigma_F}{2b^{DA}} > 0 \iff \sigma_C > \sigma_F,$$

which holds by construction of the critical/flat partition.

2.4 Two-tranche ladder under ex-ante demand uncertainty

To derive a non-trivial two-tranche ladder (not just a degenerate single-point bid at the Cournot quantity), we follow the reviewer’s suggestion and put the demand uncertainty *before* the bid: the firm commits its schedule (p_q^L, k_q^L, p_q^H) before μ_q realises. After posting, $\mu_q \sim \text{Uniform}[-\Delta_\mu, +\Delta_\mu]$ is drawn and the market clears at the marginal tranche. The hour-class assumption $\Delta_\mu \in \{\Delta_C, \Delta_F\}$ with $\Delta_C > \Delta_F$ enters here through the support of μ_q .

State-contingent profit. For a realised μ_q , residual demand at price p is $q(p, \mu_q) = (\bar{A} + \mu_q) - b^{DA}p$. With two-tranche bid $(p_q^L, k_q^L), (p_q^H, K - k_q^L)$, the firm’s profit in each state is determined by which tranche is marginal. Define the thresholds

$$\mu^* \equiv k_q^L + b^{DA}p_q^L - \bar{A}, \quad \mu^{**} \equiv k_q^L + b^{DA}p_q^H - \bar{A}, \quad (9)$$

with $\mu^{**} - \mu^* = b^{DA} \Delta p$. For $\mu_q < \mu^*$ the low tranche is marginal at price p_q^L ; the firm sells $q^L = \bar{A} + \mu_q - b^{DA}p_q^L$ at p_q^L , with profit $(p_q^L - c)q^L$. For $\mu_q > \mu^{**}$ the high tranche is marginal at p_q^H ; the firm sells $q^H = \bar{A} + \mu_q - b^{DA}p_q^H$ at p_q^H , with profit $(p_q^H - c)q^H$. For $\mu_q \in [\mu^*, \mu^{**}]$ the clearing intersection lies on the vertical capacity step at quantity k_q^L ; clearing falls at price $p_q = (\bar{A} + \mu_q - k_q^L)/b^{DA} \in [p_q^L, p_q^H]$ under uniform pricing.

Simplification: pay-as-bid clearing. The two-state treatment that follows — p_q^L marginal when $\mu_q < \mu^*$, p_q^H marginal when $\mu_q > \mu^*$ — is exact under pay-as-bid clearing. Under uniform pricing the middle region $\mu_q \in [\mu^*, \mu^{**}]$ of width $b^{DA} \Delta p$ clears at the intermediate price $(\bar{A} + \mu_q - k_q^L)/b^{DA}$ instead; we treat this as an approximation that tightens as $b^{DA} \Delta p / (2\Delta_\mu)$ shrinks (small optimal spread relative to demand half-range). The qualitative comparative statics on Δp^* in Δ_μ are the same under either clearing rule; only the level of Δp^* changes.

Expected profit and FOCs. Taking expectations over μ_q uniformly on $[-\Delta_\mu, +\Delta_\mu]$:

$$\mathbb{E}[\pi] = \frac{1}{2\Delta_\mu} \left[\int_{-\Delta_\mu}^{\mu^*} (p_q^L - c) q^L d\mu + \int_{\mu^*}^{\Delta_\mu} (p_q^H - c) q^H d\mu \right].$$

Substituting $q^L = (k_q^L - \mu^*) + \mu = \mu + k_q^L - \mu^*$ in case (a) and $q^H = \mu + k_q^L - \mu^* - b^{DA} \Delta p$ in case (b) (where $\Delta p \equiv p_q^H - p_q^L$), elementary integration yields

$$\mathbb{E}[\pi] = \frac{1}{2\Delta_\mu} \left[(p_q^L - c) (\mu^* + \Delta_\mu) \left(k_q^L - \frac{\mu^* + \Delta_\mu}{2} \right) + (p_q^H - c) (\Delta_\mu - \mu^*) \left(k_q^L - b^{DA} \Delta p + \frac{\Delta_\mu - \mu^*}{2} \right) \right]. \quad (10)$$

Taking first-order conditions in (p_q^L, p_q^H, k_q^L) , using $\partial\mu^*/\partial p_q^L = b^{DA}$, $\partial\mu^*/\partial k_q^L = 1$, and the symmetry of the uniform distribution, the system admits the closed-form interior solution

$$\boxed{p_q^{L*} = \frac{\bar{A} + b^{DA}c}{2b^{DA}} - \frac{\Delta_\mu}{4b^{DA}}, \quad p_q^{H*} = \frac{\bar{A} + b^{DA}c}{2b^{DA}} + \frac{\Delta_\mu}{4b^{DA}}, \quad \mu^{**} = 0, \quad k_q^{L*} = \frac{\bar{A} - b^{DA}c}{2} + \frac{\Delta_\mu}{4}.} \quad (11)$$

The tranche prices coincide with the conditional Cournot prices given which case applies: p_q^{L*} = Cournot at expected demand $\mathbb{E}[\mu \mid \mu < 0] = -\Delta_\mu/2$, and p_q^{H*} = Cournot at $\mathbb{E}[\mu \mid \mu > 0] = +\Delta_\mu/2$. The optimal threshold $\mu^{**} = 0$ follows from the symmetry of the uniform distribution about zero.

Comparative statics on the optimal tranche spread. From (11),

$$\boxed{\Delta p_q^* \equiv p_q^{H*} - p_q^{L*} = \frac{\Delta_\mu}{2b^{DA}}, \quad \frac{\partial \Delta p_q^*}{\partial \Delta_\mu} = \frac{1}{2b^{DA}} > 0.} \quad (12)$$

The per-curve σ_p (from (4) at the symmetric optimum $w^L = w^H = 1/2$) is therefore

$$\sigma_p^* = \frac{\Delta p_q^*}{2} = \frac{\Delta_\mu}{4b^{DA}},$$

strictly increasing in the half-range of the ex-ante demand distribution.

From per-quarter σ_p to the DiD across hour classes. The hour-class-specific support of μ_q enters (11) directly: $\Delta_\mu = \Delta_C$ in critical hours, Δ_F in flat hours. The per-curve σ_p DiD in our regression is then

$$\boxed{\text{DiD}(\sigma_p) = \sigma_p^{*,C} - \sigma_p^{*,F} = \frac{\Delta_C - \Delta_F}{4b^{DA}} > 0 \quad \text{whenever } \Delta_C > \Delta_F.} \quad (13)$$

Proposition 2 (σ_p widening in critical hours). *Under (2) and the ex-ante bidding game of §2.4, the optimal per-curve tranche spread satisfies $\Delta p^* = \Delta_\mu/(2b^{DA})$, monotone in Δ_μ . The critical-flat differential of σ_p under MTU15 is therefore $(\Delta_C - \Delta_F)/(4b^{DA}) > 0$.*

The MTU60 case. Under MTU60 the firm posts *one* hourly schedule. The relevant ex-ante distribution is the hourly aggregate $\tilde{\mu} = \frac{1}{Q} \sum_q \mu_q$, whose distribution is the Q -fold convolution of $U[-\Delta_\mu, +\Delta_\mu]$. Whatever its exact shape (it is a generalised Irwin–Hall distribution), $\tilde{\mu}$ is *strictly less dispersed* than any individual μ_q : $\text{Var}(\tilde{\mu}) = \text{Var}(\mu_q)/Q$, and the support of $\tilde{\mu}$ is properly contained in $[-\Delta_\mu, +\Delta_\mu]$. Re-running the ex-ante optimisation of §2.4 with this tighter distribution gives an optimal tranche spread

$$\Delta p^{*,\text{MTU60}} < \Delta p^{*,\text{MTU15}} = \frac{\Delta_\mu}{2b^{DA}}.$$

The magnitude of the gap depends on the precise shape of the Q -convolution and is bounded above by the SD ratio $1/\sqrt{Q}$, but the sign of the inequality holds for any non-degenerate distribution. The MTU60-to-MTU15 reform therefore strictly widens the per-curve σ_p at a given hour class, and the critical-flat differential $(\Delta_C - \Delta_F)/(4b^{DA})$ from (13) is larger under MTU15 than under MTU60.

2.5 N_{eff} widening

The optimal k_q^{L*} in (11) satisfies $k_q^{L*}/K \rightarrow 1/2$ as $\Delta_\mu \rightarrow (\bar{A} - b^{DA}c)$, making the two tranches symmetric in weight and pushing $N_{\text{eff}}^* \rightarrow 2$. As $\Delta_\mu \rightarrow 0$ the firm can collapse to a single tranche (k_q^{L*} at its boundary), pushing $N_{\text{eff}}^* \rightarrow 1$. The DiD on $\bar{N}_{\text{eff}}$ thus shares the sign of Proposition 2.

3 Pump-storage critical-hour arbitrage (closed form)

A storage agent with capacity Q_s , marginal cost zero, and within-hour energy neutrality ($\sum_q q_q^s = 0$, $q_q^s \in [-Q_s, +Q_s]$) optimises

$$\Pi_s = \sum_q p_q^{DA} q_q^s.$$

MTU15 (closed form). Storage discharges at the highest-price quarter and pumps at the lowest. Using (5) and the uniform-distribution range $[-\Delta_\mu, +\Delta_\mu]$ for μ_q :

$$\Pi_s^{\text{MTU15}} = Q_s \left(\max_q p_q^{DA} - \min_q p_q^{DA} \right) = Q_s \left(\frac{\max_q \mu_q - \min_q \mu_q}{2b^{DA}} \right) = \frac{Q_s \cdot \Delta_\mu}{b^{DA}}. \quad (14)$$

MTU60. A single hourly price gives $\max_q p = \min_q p$ mechanically: $\Pi_s^{\text{MTU60}} = 0$.

Proposition 3 (Pump-storage arbitrage scales with Δ_μ). *Under (2),*

$$\Pi_s^{\text{MTU15}}(C) - \Pi_s^{\text{MTU15}}(F) = \frac{Q_s(\Delta_C - \Delta_F)}{b^{DA}} > 0,$$

and the MTU60-to-MTU15 reform shifts storage arbitrage by $Q_s \cdot \Delta_\mu / b^{DA}$ in any hour with $\Delta_\mu > 0$.

4 Day-ahead cleared volume: imported Csereklyei–Khezzr withholding wedge

The Cournot framework with constant marginal cost gives the same total cleared quantity under MTU60 and MTU15 in our model (Cournot quantities sum to $Q(\bar{A} - b^{DA}c)/2$ either way), so the day-ahead CCGT auction-cleared scale-up of +59 GWh/day net of placebo is *not* derived endogenously here. To rationalise it we import the Csereklyei–Khezzr (2024) capacity-withholding mechanism, switching the direction of the comparison from their averaged-to-per-period to our per-period-to-averaged (the two reforms run in opposite directions).

The imported mechanism (Csereklyei–Khezzr 2024, Prop. 3). Suppose a second firm with marginal cost $c_2 > c$ caps the residual-monopolist’s price at c_2 in low-demand periods via Bertrand-style undercutting (this is the only place we deviate from Cournot: the second firm’s bid acts as a price ceiling, not a residual demand). Under per-period clearing, the firm sets $p_q^{DA} = c_2$ in low-demand quarters and captures the full residual demand at that price. Under averaged hourly clearing, Csereklyei–Khezzr show in their Proposition 3 that the firm has a strict incentive to deliberately withhold low-tranche capacity (k_q^L pushed downward) to elevate the hourly average price above c_2 . The withholding profit-loss tradeoff gives a strict inequality in their setup whenever the low-demand quantity d_- satisfies $(P - c_2)/2 \cdot d_q > (c_2 - c) \cdot d_{-q}$ for the lowest-demand quarter.

Adapted to MTU60 vs MTU15. The hourly clearing of MTU60 is the averaged-pricing case; per-quarter clearing of MTU15 is the per-period case. The reform direction (MTU60 $\rightarrow$ MTU15) removes the withholding incentive, so cleared volume rises under MTU15. The magnitude depends on $(c_2 - c)$ and the within-hour low/high demand asymmetry; these are not parameters of our model and we do not re-derive their algebra here.

Note on the mechanism switch. The Csereklyei–Khezr argument is Bertrand-undercutting (the second firm’s price ceiling is the binding constraint in low-demand states), distinct from the Cournot residual-demand interaction that drives our D_{price} and σ_p results. We import the result rather than try to unify the two micro-foundations; the two propositions are independent contributions that happen to point in the same direction.

5 Sequential DA-IDA bidding with two-tranche schedules

The DA-IDA arbitrage condition is the proposal’s (eq. 16 under DA15/ID15): under *competitive* arbitrage,

$$p_q^{DA} = p_q^{ID} \quad \forall q. \quad (15)$$

Under *strategic* arbitrage, the proposal’s eq. 17 gives a persistent wedge:

$$p_q^{ID} - p_q^{DA} = -\frac{p_q^{DA} - c}{4}. \quad (16)$$

Substituting our closed-form p_q^{DA} from (5):

$$p_q^{ID*} = \frac{3p_q^{DA} + c}{4} = \frac{3(\bar{A} + b^{DA}c) + 6\mu_q + 4b^{DA}c}{8b^{DA}}.$$

The IDA price inherits the cross-quarter variation of the DA price scaled by 3/4, so the cross-quarter SD of the IDA price under MTU15 is

$$D_{\text{price}}^{ID} = \frac{3\sigma_\mu}{8b^{DA}} = \frac{3}{4} D_{\text{price}}^{DA}.$$

The CCGT marginal-cost regime in IDA. The descriptive analysis documents that CCGT bids in IDA with $\sim 0\%$ withholding, i.e. *no scarcity tier* (no parked capacity above the price ceiling, in contrast to the DA-side two-tier curve). This does *not* mean the IDA ladder collapses to a single point at marginal cost: the firm still posts a non-degenerate ladder *centred at* MC, reflecting unit heterogeneity within the firm’s CCGT fleet and the same ex-ante demand uncertainty that drives §2.4 on the DA side.

The proper IDA reading is therefore symmetric to DA on the bid-shape side: each unit posts a two-tranche ladder around MC with the optimal spread of Proposition 2, and the per-curve σ_p^{IDA} inherits the same closed form $\Delta_\mu/(4b^{DA})$ in critical-hour quarters. The empirical ID15 DiD of +2.45*** for CCGT thus has the same derivation as the DA-side prediction. The *absolute level* of IDA bids differs from DA bids because the ladder is centred at c rather than at the Cournot mark-up $(\bar{A} + b^{ID}c)/(2b^{ID})$, but the *spread* comparative static is the same. On the price-level side, the cross-quarter D_{price}^{IDA} inherits the DA Cournot variation through the arbitrage condition above (scaled by 3/4 under the strategic wedge (16)).

6 Mapping to empirical findings

Empirical finding	Closed form	Equation
D_{price} widens in critical hours under MTU15	$\sigma_{\mu}/(2b^{DA})$	(8), Prop. 1
σ_p widens in critical hours under MTU15 (DiD)	$(\Delta_C - \Delta_F)/(4b^{DA})$	(13), Prop. 2
N_{eff} widens in critical hours under MTU15	corollary of Prop. 2	
Pump-storage critical-hour discharge scales up	$Q_s \Delta_{\mu}/b^{DA}$	(14), Prop. 3
DA15 CCGT auction-cleared +59 GWh/day net	imported from Csereklyei–Khezr (2024)	§4
ID15 IDA price drop (directional)	$3p_q^{DA}/4 + c/4$	(16)

7 Discussion: what the closed form delivers and what it does not

Scope of the model. The model’s purpose is to illustrate the granularity-to-bid-shape mechanism in closed form, not to provide quantitative predictions for calibration or out-of-sample testing. The “what is not delivered” list below should be read as appropriate scope-setting rather than missing pieces.

What is delivered. Closed-form solutions for all per-quarter clearing prices, per-tranche bids, storage arbitrage, and the DiD predictions on D_{price} , σ_p , N_{eff} and storage discharge. All comparative statics are direct algebra under the uniform-distribution assumption (2), and the proofs are elementary substitutions.

What is not delivered. (i) The strategic mark-up channel beyond the linear-Cournot baseline (e.g., scarcity-tier rents from the two-tier CCGT bid documented descriptively) requires departing from constant marginal cost. (ii) The two-tranche derivation in §2.4 simplifies away the middle vertical-step region of width $b^{DA}\Delta p^*$ where clearing falls on the firm’s capacity step; this is exact under pay-as-bid clearing but an approximation under uniform pricing. (iii) The Csereklyei–Khezr withholding wedge is reproduced verbatim from their Proposition 3; the magnitude depends on the relative cost gap ($c_2 - c$) and the within-hour low/high demand asymmetry, both of which are taken from their notation.

Numerical magnitudes are not the point. The model is designed to deliver signs and comparative statics. The mapping table in §6 gives the closed-form proportionality for each predicted direction; we do not assign numerical values to $(\bar{A}, b^{DA}, c, \Delta_C, \Delta_F)$ because the empirical identification is in the differential coefficients, not in the level values that the parameters would pin down.

References.

- Csereklyei, Z. and P. Khezr (2024), “How do changes in settlement periods affect wholesale market prices? Evidence from Australia’s National Electricity Market,” *Energy Economics* 132, 107425.
- Ito, K. and M. Reguant (2016), “Sequential markets, market power, and arbitrage,” *American Economic Review* 106(7), 1921–1957.